

higher-dimensional spaces, subtle notions of equivalence, and category theory structures. Their blockage on mental arithmetic becomes irrelevant.

It seems to me that we are denying students entry into abstract mathematics when they struggle with non-abstract mathematics, and that this approach is counter-productive. Or perhaps some students self-select out of abstract mathematics if they did not enjoy non-abstract mathematics. This is as if we didn't let people try swimming because they are a slow runner, or if we didn't let them sing until they're good at the piano.

One aim of this book is to present abstract mathematics directly, in a way that does not depend on proficiency with other parts of mathematics. It doesn't have to matter if you didn't make it over some of those earlier hurdles.

Traditional mathematics: methods

When I studied modern languages at school there were four facets tested with different exams: reading, writing, speaking and listening. Of those, writing and speaking are “productive” where reading and listening are “receptive”. For full mastery of the language all four are needed of course, but if complete fluency is beyond you it can still be rewarding to be able to do only some of these things. I later studied German for the purposes of understanding the songs of Schubert (and Brahms, Strauss, Schumann, and so on). My productive German is almost non-existent, but I can understand Romantic German poetry at a level including some nuance, and this is rewarding for me and helps me in my life as a collaborative pianist.

I think there is a notion of “productive” and “receptive” mathematics as well. Productive mathematics is about being able to answer questions, say, homework questions or exam questions, and, later on, produce original research. There is a fairly widely held view that the only way to understand math is to work through problems. There is a further view that this is the only way of doing math that is worthwhile. I would like to change that.

I view “receptive” mathematics as being about appreciating math even if you can't solve unseen problems. It's being able to follow an argument even if you wouldn't be able to build it yourself.

I can appreciate German poetry, restaurant food, a violin concerto, a Caravaggio, a tennis match. Imagine if appreciation were only taught by doing. I can even read a medical research paper although I can't practice medicine. The former is still valuable. In math some authors call this “mathematical tourism” with undertones of disdain. But I think tourism is fine — it would be a shame if the only options for traveling were to move somewhere to live there or else